

Arun Kumar Nandam

Master's in Computer Science, Illinois Institute of Technology

Madison, WI | +1 (872) 239-8074 | kumar.nandam29@gmail.com | USA | [Linkedin](#)

PROFESSIONAL SUMMARY

- Experienced Data Engineer with around 5 years of experience in designing and managing large-scale data systems on cloud computing providers (**AWS, Azure, and GCP**).
- Advanced proficiency in **SQL** for querying relational database and NoSQL databases, including **MySQL, PostgreSQL, MongoDB, and Cassandra**, with strong skills in **Python, Scala** for building and automating data pipelines.
- Skilled in **big data processing** with **Apache Spark, Hadoop**, and Enterprise data warehousing solutions like **AWS Redshift, Synapse, Snowflake, and BigQuery** for analytics.
- Proficient in designing and deploying serverless architectures using **AWS Lambda**, building ETL pipelines with **AWS Glue**, and enabling real-time data streaming with **AWS Kinesis, DynamoDB, and S3**.
- Skilled in **Azure** services such as **Data Factory, Databricks, Synapse Analytics**, and **Blob Storage** for data integration, transformation, and analytics to streamline workflows and enable actionable insights.
- Proficient in **CI/CD pipelines** (Jenkins, Git, Terraform), **containerized applications** (Docker, Kubernetes, Helm), and **API development**, ensuring seamless deployment, performance, scalability, and reliability.
- Experienced with data visualization tools like **Tableau** and **Power BI** for creating interactive dashboards and reporting.
- Strong **cross-functional collaboration** with business teams, data stewards, and stakeholders for **data governance and quality assurance**.
- Strong communication skills – written, verbal with keen attention to detail, coupled with strong analytical and problem-solving skills, ability to work in fast-paced, collaborative environment and to drive teamwork and project success.

WORK EXPERIENCE

Data Engineer

Jan 2024 – Present

Community Health Systems, Franklin, Tennessee, USA

- Designed **Azure Data Factory (ADF)** ETL pipelines to transform healthcare data and orchestrated end-to-end workflows using Apache Airflow, automating job scheduling, and **improving data processing efficiency by 50%**.
- Processed genomic data via **Spark** on **Databricks** and used **Snowflake** as the analytical DWH layer, improving downstream query performance for diagnostics, leveraging **Terraform** to automate infrastructure provisioning.
- Migrated healthcare data (HL7 v2/v3, X12, FHIR, JSON) from on-prem to **ADLS** and **Snowflake using Informatica PowerCenter**, leveraging **dbt** for ELT model builds and version control to improve data accessibility.
- Architected healthcare **data warehouse models** and data marts in **Snowflake** following **Kimball modeling** principles and leveraged **dbt** to version, document, and test ELT models supporting ML analytics and advanced analytics.
- Developed **Azure Functions** for serverless data processing and event-driven automation, processed analytics in **Snowflake SQL**, replacing Synapse clusters for improved concurrency and reporting on large-scale healthcare datasets.
- Integrated **Azure DevOps** with **Snowflake** with **Azure** for advanced analytics and secure data sharing. Built real-time fault-tolerant streaming pipelines using **Kafka** on **Azure Event Hubs**, enabling scalable ingestion of patient records.
- Built a data governance framework using **Azure Purview, Azure Data Catalog, and Python**, automating data quality checks, lineage tracking, ensuring **HIPAA** security compliance for patient records stored in Azure Data Lake.
- Integrated **Power BI with Snowflake, Synapse, and PostgreSQL**, using **DAX** calculations to reduce healthcare reporting time by 40% and deliver actionable insights to technical and non-technical stakeholders.

AWS Data Engineer

May 2023 – Dec 2023

Nissan North America, Franklin, Tennessee, USA

- Developed scalable data pipelines and ETL workflows using **AWS Glue, Lambda, and Step Functions**, processing 2TB+ of manufacturing data daily and **improving efficiency by 35%**. Built and optimized data warehouses and data lakes using **Amazon Redshift, S3, and Athena** enhancing analytics capabilities for business processes.
- Work collaboratively with cross-functional teams such as data analyst, software engineers, data scientists, leveraging **agile/scrum methodologies** to **dbt** models, end-user documentation, sprint planning, and process improvements.
- Designed monitoring solutions with **AWS CloudWatch, X-Ray, and Elasticsearch**, achieving 99.9% system uptime.

- Automated CI/CD strategic workflows with **Jenkins** and **AWS CodePipeline**, integrating **dbt model** builds and tests into the deployment lifecycle to ensure reliable, production-ready data transformations.
- Built batch & real-time data pipelines with **Apache Spark**, **Kafka**, **AWS Kinesis** and **AWS EMR** for high-velocity signal processing and developed API integrations using **AWS Lambda**, **DynamoDB** to connect IoT analytics systems.
- Deployed containerized applications using **Docker**, **ECS**, **EKS**, and **Kubernetes**, collaborating with product teams to build and deploy machine learning models on **SageMaker**, achieving a 15% growth in predictive accuracy.

GCP Data Engineer

Nov 2020 – Dec 2022

Big Lots, Inc., Bangalore, India

- Designed and implemented scalable data pipelines using **Google Cloud Dataflow**, **Apache Beam**, and used **dbt** in **BigQuery** to build modular **SQL models** for retail and inventory data, improving transformation efficiency by 30%.
- Built and optimized data lakes on **Google Cloud Storage (GCS)** for structured, semi-structured, and unstructured data.
- Automated **ETL** workflows with **Google Cloud Composer (Airflow)**, transforming data across 20+ retail systems and orchestrating dbt runs for scheduled model builds and data quality tests across systems.
- Deployed and managed **DataProc** clusters with **PySpark** and **Hadoop** for large-scale batch processing including statistical methods, streamlining retail workflow and configured **Pub/Sub** for real-time streaming data ingestion from POS devices and e-commerce platforms, enabling seamless integration.
- Implemented **IAM roles**, fine-grained access controls, and **VPC Service Controls** securing sensitive customer and sales data, ensuring 100% compliance with data protection policies.
- Built **continuous integration of pipelines** using **Cloud Build**, **Jenkins**, and **GitLab CI**, reducing deployment times by 40% and ensuring scalability of reliable workflows.
- Designed dashboards with **Tableau** for sales trends and inventory forecasting, enabling data-driven decision-making.
- Assisted in the development of **data governance** best practices, classifications, and metadata management, ensuring compliance with security, data protection policies, and improving data quality across manufacturing data solutions.

Data Engineer - Fintech

Sep 2019 – Oct 2020

Synchrony Financial, Bangalore, India

- Developed and maintained scalable data pipelines using **Apache Spark**, **Databricks**, and **Kafka**, processing 1TB+ of real-time data daily for customer insights and operational efficiency.
- Implemented ETL workflows with **Apache Airflow** and **Azure Data Factory**, streamlining data ingestion and transformation across 10+ source systems and Automate processes, reducing manual intervention by 50%.
- Optimized SQL Server(T-SQL) performance for large-scale processing, integrating SSIS for efficient data transformation. Automated SSRS reporting to deliver real-time insights, reducing manual efforts, and enhancing business intelligence.
- Built and optimized data lakes on **Azure Data Lake Storage** to efficiently store and manage structured, semi-structured (XML, JSON, Parquet), and unstructured data and designed high-performance star schema data warehouses with **Azure Synapse** and **Snowflake**, improving query performance by 40% and enabling advanced analytics and reporting.
- Developed real-time innovative **Power BI dashboards** on top of **dbt-modeled Snowflake** tables, implementing DAX calculations and optimizing back-end processes to deliver actionable insights on customer engagement.
- Defined and implemented key performance metrics (KPIs) and success data flows metrics to measure data pipeline efficiency, **reducing ETL failures by 30% and** improving business data driven decisions, finance and performance.

SKILLS

AWS: S3, Redshift, Glue, Lambda, EC2, Kinesis, RDS, Dynamo DB, SQS, SNS, EMR

Azure: Data Factory, Data Lake, Synapse Analytics, Databricks, Data Studio, Azure SQL, Azure ML

Machine Learning: decision trees, A/B Testing, Statistical modeling (Regression, Classification, clustering), NLP, AI/ML

ETL/ELT: Apache Airflow, Informatica, IBM DataStage, Apache Nifi, Apache Kafka, Splunk, Apache Flink, dbt, Fivetran

Programming languages: Python (Pandas, NumPy, Flask, Scikit-learn, TensorFlow, Keras), Java, SQL, R, Statistics

Distributed systems: Apache Spark (PySpark), big data technologies (Hadoop, Pig, Hive, HBase, Presto, Sqoop)

Databases: Oracle, PL/SQL, MySQL (CTEs, OLAP, OLTP), Postgres, Snowflake, Teradata, SSAS, MongoDB, Cassandra

Data Visualization software: Streamlit, Tableau, MS Power BI, SSRS

DevOps practices: Terraform, version control system (Git, GitHub), Containerization (Docker, Kubernetes), REST API

Additional Skills: Unix, JIRA, MS Excel, Shell Scripting, Information Technology, Confluence, Google BigQuery

CERTIFICATIONS - Google Cloud Associate Data Engineer, Azure Data Fundamentals